

LET Sample Examination 2020 (2025) – Marking Guide**Q.1 – Should (2 Marks)****Clause 1.4.108 (2 Marks)****Q.2 – Suitable for the conditions (2 Marks)****Clause 3.3.2.7 (2 Marks)****Q.3 – To provide an earth reference (2 Marks)****Clause 4.14.5 Exception (2 Marks)****Q.4 – 10A (2 Marks) (deduct 1 mark for no or incorrect units)****Clause 3.6.2 Exception 1 (2 Marks)****Q.5 – When suitably protected against mechanical damage (2 Marks)****Clause 2.5.5 (2 Marks)****Q.6 – Accept either of the following**

6 Months (2 Marks)

Clause 3.3.2 Table 6 (2 Marks)**3.2 (d) Table 6 (check?)****Q.7 – High Voltage****Clause 105 (2 Marks)****Q.8 – 1 – An ambulance should be called (2 Marks) 2 – CPR started (2 Marks)****Q.9**Table 3(3) Item 4 Table 14 Col 23 - $35\text{mm}^2 = 172\text{A}$ (1 Mark) *(Item number is optional do not deduct marks)**Part (i)* $220/2 = 110\text{A}$ (1 Marks)

Table 25(2) Col 2 Derating for more than one circuit = 0.81 (1 Mark)

Table 28(1) Col 2 Derating for Depth 1.0m = 0.95 (1 Mark)

 $344 \times 0.81 \times 0.95 = 264.70\text{A}$ (1 Mark) $= 25\text{mm}^2$ (1 Marks)*Part (ii)*(25mm² just, but after HRC derate.....)

HRC Derate = 0.9 (1 Mark)

 $= 238.23\text{A}$ (1 Mark)**(Deduct 1 Mark for no or incorrect units)**

Q.10**(2 marks for correct defect one mark for the correct clause)****(Only accept the first 4 defects candidate has listed)**

1. Consumers' mains are not installed in a manner as to be capable of maintaining supply when exposed to fire **7.2.5.2.1 (a)** **7.2.2.1**
2. The cable to the fire pump is undersized – **3.4.1**
3. Main switch distribution board does not indicate the on/off position – **2.3.2.2.1(c)**
4. The main neutral connection at the neutral bar is not labelled – **2.10.5.4**
5. The telecommunications earthing conductor is undersized, 6mm² minimum **-5.6.2.7 (iv)**
6. The Main earth conductor is not insulated above the ground- **5.3.2.4**
7. The Main Switch Distribution Board is not clearly marked to indicate what portion of the Installation it controls- **2.3.3.5. (b)**
8. Main Switch Fire Pump not labelled "MAIN SWITCH" in uppercase – **7.2.4.4 (a)**
9. Main Switch Fire Pump not labelled "IN THE EVENT OF FIRE DO NOT SWITCH OFF" in uppercase – **7.2.4.4 (b)**
10. No date of initial certification on SWB – **8.4**

Q.11

Meter X – 2200W (3 Marks)

Meter Y – 2A (3 Marks)

Meter Z – 440V (3 Marks)

(Deduct 1 Mark for no or incorrect units)**Q.12****Table C2 Column 2 (1 Mark)**

Equipment	Load Group	Calculation	Maximum Demand
10A Socket outlets 36 Points total	B(ii)	$1000 + (35 \times 100) = 4500 / 230 = 19.57A$	19.57A (2 Marks)
20 A Socket Outlets 4 Points total	B(iii)	$20 + ((3 \times 20) \times 0.5) = 50A$	50A (1 Mark)
Lights 36 Points Total	A	$((36 \times 60) / 230) \times 0.75 = 7.04A$	7.04A (1 Mark)
8.2KW Range 6.2KW Split System	C C	FCL of highest rated then 50% of the remainder. $8200 / 230 = 35.65A$ $((6200 / 230) \times 0.5) = 13.48A$	35.65A (1 Mark) 13.48A (1Mark)
		Total Maximum Demand	125.74A (1 Marks)

(Deduct 1 Mark for no or incorrect units)

Q.13**Consumers Mains**

Table 41 Col. 10 $V_c = 0.491$ $V_d = 0.61v$ (3 Marks)

Sub-mains

Table 41 Col. 10 $V_c = 4.30$ $V_d = 5.81v$ (3 Marks)

Final sub-circuit

Table 42 Col. 6 $V_c = 6.49$ $V_d = 6.02v$ (3 Marks)

Total Voltage Drop

$0.61 + 5.81 + 6.02 = 12.44v$ (1 Mark)

(Deduct 1 Mark for no or incorrect units. Deduct 1 mark for no or incorrect table number)

Q.14

Ratio = 4:1 (1 Mark) (deduct 1 Mark if unit is given)

Minimum Time = Accept 2-2.5 seconds (1 Mark)

Maximum Time = Accept 7-9 seconds (1 Mark)

(Deduct 1 mark if unit is given for ratio or unit for no or incorrect time unit)

Q.15

Transformer impedance = 0.01644Ω (3 Marks)

MSB Fault Current = 9410.80A (3 Marks)

DB Fault Current – 2756.47A (3 Marks)

(Deduct 1 Mark from each answer if candidate has rounded transformer impedance to 4 decimal places instead of 5)

(Deduct 1 Mark for no or incorrect units)

Q.16

(a) The maximum demand of the portion of the electrical installation being protected by the device.

(b) The highest current rating of any overload protective device on the portion of the electrical installation being protected.

Clause number – 2.6.2.1 (a) (b) (3 Marks)

Q.17

B (2 Marks)

Q.18

1. 25 (2 Marks)

Clause number – 2.3.4 (2 Marks)